Problem 8.6

Calculate the accumulated value at the end of 3 years of 15,000 payable now assuming an interest rate equivalent to an annual discount rate of 8%.

Solution

Let $d = 0.08$ be the annual effective discount rate. The equivalent effective interest rate i satisfies

$$d = \frac{i}{1+i}.$$

So,

$$0.08 = \frac{i}{1+i}.$$

Multiplying both sides by $(1+i)$ gives

$$0.08 + 0.08i = i.$$

Hence,

$$0.08 = 0.92i, \qquad i = \frac{0.08}{0.92} = \frac{2}{23} \approx 0.0869565.$$

Now accumulate 15,000 for 3 years at rate i:

$$15{,}000(1+i)^3 = 15{,}000\left(1 + \frac{2}{23}\right)^3.$$

Thus,

$$15{,}000(1.0869565)^3 \approx 19{,}263.17.$$

Therefore, the accumulated value at the end of 3 years is

$$\boxed{19{,}263.17}.$$